Lecture 11

Properties of expectation, variance and moments

Properties of expectation

① If $Y = aX + b$, where a and b are finite constants, then

$$E(Y) = a\,E(X) + b.$$

Example $E(2X - 7) = 2E(X) - 7.$

② $E(a_1 X_1 + a_2 X_2 + \ldots + a_n X_n) =$

$$a_1 E(X_1) + a_2 E(X_2) + \ldots + a_n E(X_n),$$

if $a_1, \ldots, a_n$ are all constants.

IMPORTANT ② holds for ANY random variables $X_1, \ldots, X_n$.

Example (The expectation of $X \sim Bin(n,p)$)

- Recall that
  $X = X_1 + \dots + X_n$, where $X_1, \dots, X_n$ are independent and each $X_i \begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix}$.

- Then

$$E(X) = \sum_{i=1}^{n} E(X_i)$$
$$= \sum_{i=1}^{n} \left(1 \cdot p + 0 \cdot (1-p)\right) = np.$$

Note A direct calculation of

$$E(X) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$$ is

possible, but more difficult.

Conclusion: If $X \sim Bin(n,p)$, then $E(X) = np$.

③ If $X_1, X_2, \ldots, X_n$ are **independent** random variables, then:

$$E(X_1 \cdot X_2 \cdot \ldots \cdot X_n) = E(X_1) \cdot E(X_2) \cdot \ldots \cdot E(X_n)$$

Example Assume that X_1, X_2 and X_3 are independent and assume that:

$$E(X_1) = E(X_2) = E(X_3) = 0$$
$$E(X_1^2) = E(X_2^2) = E(X_3^2) = 1.$$

Calculate $\circledast$ $E\left[X_2^2 \cdot (X_1 - 3X_3)^2\right]$.

Solution

$$\circledast \overset{\text{Independence}}{=} E(X_2^2) \cdot E\left[(X_1 - 3X_3)\right]^2$$

$$= 1 \cdot E(X_1^2 + 9X_3^2 - 6X_1X_3)$$

$$= E(X_1^2) + 9E(X_3^2) - 6E(X_1) \cdot E(X_3)$$

$$= 1 + 9 - 6 \cdot 0 \cdot 0 = 10.$$

When equality does not hold

(1) $E\left(\prod_{i=1}^{n} X_i\right)$ is NOT necessarily equal to $\prod_{i=1}^{n} E(X_i)$, if the X_i's are not independent.

(2) • If g is some arbitrary function, $E(g(X))$ is typically NOT EQUAL TO $g(E(g(X)))$.

See Example 4.1.13, with $g(x) = \sqrt{x}$.

• For g = linear, equality holds.

• For g = convex function, we always have $E(g(X)) \geq g(E(X))$ (Jensen's inequality!)

The variance of a random variable

Motivation

- Consider $X \sim f(x) = \begin{cases} 2x, & \text{for } 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$

$E(X) = \frac{2}{3}$

- Consider $Y = \frac{2}{3}$, the constant random variable always equal to $\frac{2}{3}$. $E(Y) = \frac{2}{3}$.

- X and Y have the same mean, but very different distributions.

- How to characterize how <u>spread out</u> a distribution is? One possibility: via <u>the variance</u> of a distribution.

Definition If X is a r.v. and $E(X) = \mu$, then

$$\text{var}(X) =: E(X-\mu)^2.$$

↑ "Variance of X"

. The standard deviation of X is $\sqrt{\text{var}(X)}$.

Result (Useful way of calculating the variance)

$$\text{var}(X) = E(X^2) - [E(X)]^2.$$

Properties ① $\text{var}(X) \geq 0$.

② $\text{var}(X) = 0$ iff $(\exists)$ a constant c such that $\mathbb{P}(X = c) = 1$.

③ $\text{var}(aX + b) = a^2 \cdot \text{var}(X)$.

④ If $X_1, \ldots, X_n$ are <u>INDEPENDENT</u>, then

$$\text{var}(X_1 + X_2 + \ldots + X_n) = \text{var}(X_1) + \ldots + \text{var}(X_n).$$

$$\text{var}(a_1 X_1 + \ldots + a_n X_n) = a_1^2 \cdot \text{var}(X_1) + \ldots + a_n^2 \cdot \text{var}(X_n)$$

<u>What if the variance does not exist ?</u>
<u>How to measure the spread of the distribution in alternative ways</u>

- If X has a Cauchy distribution, then $E(X)$ does not exist, therefore the variance(X) does not exist.

<u>Definition</u> Interquartile range (IQR)

$$IQR =: F^{-1}(0.75) - F^{-1}(0.25),$$ and is an alternative measure of the spread of a distribution.

- The variance of $X \sim Bin(n,p)$

Since $X = X_1 + \dots + X_n$ defined at page (1'),

$$var(X) = \sum_{i=1}^{n} var(X_i);$$

$$X_i \begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix}; \quad X_i^2 \begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix}$$

$$var(X_i) = E(X_i)^2 - [E(X_i)]^2$$

$$= p - p^2 = p(1-p).$$

Therefore

if $X \sim Bin(n,p)$,

$var(X) = n \cdot p(1-p)$

The mean and the median of a distribution

Definition Let X be a random variable.

Every number m such that

$\mathbb{P}(X \geq m) \geq \frac{1}{2}$ *and* $\mathbb{P}(X \leq m) \geq \frac{1}{2}$ is

called a *median*.

Remarks

- Every distribution has *at least* one median.
- Many distributions have multiple medians.
- The 50% quantile is *a* median.

→ Examples 4.5.1 — 4.5.4.

- **Result** If X is a r.v. taking values in an interval I, and m is the median of X, then $h(m)$ is also a median of $h(X)$, if h is a one-to-one function.

(**) Recall that this is *NOT* true for the mean.

Example 4.5.1 The Median of a Discrete Distribution. Suppose that X has the following discrete distribution:

$$\Pr(X = 1) = 0.1, \qquad \Pr(X = 2) = 0.2,$$
$$\Pr(X = 3) = 0.3, \qquad \Pr(X = 4) = 0.4.$$

The value 3 is a median of this distribution because $\Pr(X \le 3) = 0.6$, which is greater than 1/2, and $\Pr(X \ge 3) = 0.7$, which is also greater than 1/2. Furthermore, 3 is the unique median of this distribution. ◀

Example 4.5.2 A Discrete Distribution for Which the Median Is Not Unique. Suppose that X has the following discrete distribution:

$$\Pr(X = 1) = 0.1, \qquad \Pr(X = 2) = 0.4,$$
$$\Pr(X = 3) = 0.3, \qquad \Pr(X = 4) = 0.2.$$

Here, $\Pr(X \le 2) = 1/2$, and $\Pr(X \ge 3) = 1/2$. Therefore, every value of m in the closed interval $2 \le m \le 3$ will be a median of this distribution. The most popular choice of median of this distribution would be the midpoint 2.5. ◀

Example 4.5.3 The Median of a Continuous Distribution. Suppose that X has a continuous distribution for which the p.d.f. is as follows:

$$f(x) = \begin{cases} 4x^3 & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

The unique median of this distribution will be the number m such that

$$\int_0^m 4x^3\,dx = \int_m^1 4x^3\,dx = \frac{1}{2}.$$

This number is $m = 1/2^{1/4}$. ◀

Example 4.5.4 A Continuous Distribution for Which the Median Is Not Unique. Suppose that X has a continuous distribution for which the p.d.f. is as follows:

$$f(x) = \begin{cases} 1/2 & \text{for } 0 \le x \le 1, \\ 1 & \text{for } 2.5 \le x \le 3, \\ 0 & \text{otherwise.} \end{cases}$$

Here, for every value of m in the closed interval $1 \le m \le 2.5$, $\Pr(X \le m) = \Pr(X \ge m) = 1/2$. Therefore, every value of m in the interval $1 \le m \le 2.5$ is a median of this distribution. ◀

Prediction (How to use the mean and/or median).

- Let X be a r.v. and let

① $\begin{cases} E(X) = \mu \\ \text{Var}(X) = \sigma^2. \end{cases}$; ② $\text{Median}(X) = m$.

- Values of X will be observed in an "experiment", but we would like to predict a value <u>before</u> the experiment, when we have access to information ④.

- We want to make the smallest possible <u>error</u> when we predict.
- How to define error? There are many possibilities!
- <u>Definition</u> (Mean squared error/MSE) MSE =: $E[(X-d)^2]$, the MSE of predicting that a value of X is $\underline{\underline{d}}$.
- <u>Theorem</u> $E[(X-\mu)^2] \leq E[(X-d)]^2$, for [any] d, with equality iff $\mu = d$.

<u>Moral</u> Predicting that the "next" value of X is $\mu = E(X)$ gives the smallest MSE.

- We can also define the prediction error as below:

Definition 2 (Mean absolute error (MAE))

$E(|X-d|)$ is called the MAE for predicting d as a value of X.

Theorem If median $(X) = m$, then

$E(|X-m|) \leq E(|X-d|)$, for any d, with equality iff $d = m$.

- See the examples 4.5.7, 4.5.8 and 4.5.9 for mean/median calculations.

Example 4.5.9 Predicting a Discrete Uniform Random Variable. Suppose that the probability is 1/6 that a random variable X will take each of the following six values: 1, 2, 3, 4, 5, 6. We shall determine the prediction for which the M.S.E. is minimum and the prediction for which the M.A.E. is minimum.

In this example,

$$E(X) = \frac{1}{6}(1+2+3+4+5+6) = 3.5.$$

Therefore, the M.S.E. will be minimized by the unique value $d = 3.5$.

Also, every number m in the closed interval $3 \leq m \leq 4$ is a median of the given distribution. Therefore, the M.A.E. will be minimized by every value of d such that $3 \leq d \leq 4$ and only by such a value of d. Because the distribution of X is symmetric, the mean of X is also a median of X. ◀